

Spectral Graph Power Dynamics

IAS2026 closure poster draft | scoped claim with negative transfer evidence

QUESTION	SURVIVED	TRANSFER RESULT
Does the frozen IEEE-39 benchmark admit a compact spectral/collective closure explanation of the P4 blocker, and does it transfer to the archived V9 portfolio set?	P4 H4 nominal alpha	Historical V9 reveal
	0.1270064667 s^-1	395/512 correct
METHOD - Frozen-artifact Gate 0 reproduction - Randomized theorem/property tests - Retrospective F7A and G2 nonlinear audit - Independent PowerDynamics tutorial equilibrium gate LABELS IEEE39_VALIDATED THEOREM_VALIDATED NONLINEAR_TDS_VALIDATED POWERDYNAMICS_VALIDATED	P4 proper stable subsets: 15 Boundary witness: g*=0.2076814, f=0.7064248 Hz Frozen TDS agreement: 32/32 Governor policy: 0.1270065 -> -0.074541 s^-1 THEOREM CHECK Four randomized test modules pass, with the contextual-return assumption Q_ii=0 explicit.	116 false-safe; 1 false-unstable The stronger 511/512, zero-error expectation is REFUTED . DO NOT OVERCLAIM - No universal IEEE-39 claim - No physical robust radius - No second converter model - No new blind holdout - No same-mechanism PowerDynamics parity - No EMT/current-limit/DC-link certification

SAFE POSTER CLAIM: In the frozen IEEE-39 benchmark and stated policy, the spectral/collective closure audit reproduces a P4 minimal incompatibility and the declared frozen nonlinear verdicts. The historical V9 transfer expectation is refuted, the documented governor changes the P4 H4 sign, and robustness, second-model, same-mechanism PowerDynamics, and new blind-holdout claims remain untested or stopped by scope.